6.12 Function Practice (Lab 6.0)

LAB ACTIVITY | 6.12.1: Function Practice (Lab 6.0) | Full screen | 0 / 10

First, download this template and rename it `function_practice.py`:

- template.py

Do the following exercises:

1. Write a function, `compute_cost`, that receives `miles`, `miles_per_gallon`, and `dollars_per_gallon` for a road trip, and returns the cost in dollars to drive those miles. For example, if the function is called with: `compute_cost(200, 20, 4)` it returns: 40.0, i.e., a 200-mile trip in a car that gets 20 miles per gallon where the gasoline costs $4 per gallon. You may assume that the miles/gallon value is not 0. As a starting point, here is a non-functional implementation of the computation:

```
(miles / miles_per_gallon) * dollars_per_gallon
```

2. Write a function, `count_spaces`, that receives a string, `s`, and returns the number of spaces in that string. For example, the function call `count_spaces('one two')` returns 1. As a starting point, here is a code segment (not a function) that implements this counting (do not use a pre-built function (e.g., `my_string.count(" ")`):

```
# This code segment assumes that some_string has a value (e.g., 'one two').
count = 0
for c in some_string:
    if c == ' ':
        count += 1
# Now, count's value should be the number of spaces in the string (e.g., 1).
```

This sort of algorithm is commonly called an *accumulator* because it accumulates a count of the spaces.

Bundle these two code segments into function definitions as appropriate. Remember to include a docstring for each function implement at least three appropriate test cases for each using `assert` before submitting it.

Grading

- **90% — Correctness**: Your programs should run as specified, with appropriate testing.
- **10% — Understandability**: Your programs should be easy to read.

*These exercises are based on ZyLabs examples.

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Submit for grading

Coding trail of your work What is this?

```
History of your effort will appear here once you begin working on this zyLab.
```

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